

Chronic diseases such as atherosclerosis and psoriasis are associated with alterations to the lipid composition of cell membranes. Atherosclerosis is caused by the accumulation of cholesterol in the endothelial cells lining the arterial wall. Cholesterol carried by low-density lipoprotein (LDL) is deposited into cell membranes, where it forms cholesterol crystalline domains (CCDs). In damaged cells, these CCDs increase membrane permeability and contribute to the formation of cholesterol plaques that can potentially block the artery itself. Long-chain omega-3 fatty acids (O3FAs) derived from marine animals may slow the development of atherosclerosis by preventing cholesterol from organizing into CCDs. Different formulations of O3FA typically contain varying concentrations of eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) (Figure 1).

Atherosclerosis is commonly treated by administering drugs called statins that decrease the free cholesterol in blood plasma by inhibiting the production of a cholesterol precursor. Statin therapy combined with O3FAs supplementation may be used to reduce atherosclerosis in patients with persistent hypertriglyceridemia (high triglyceride plasma levels).

Psoriasis is a skin disease in which the permeability of the uppermost layer of the epidermis, the stratum corneum (SC), is compromised due to a significant decrease in the level of specialized lipids called ceramides. The scarcity of ceramides impedes the formation of a lipid matrix called the long periodicity phase in the cell membrane and can lead to water loss. Increasing the concentration of ceramides such as esterified omega-hydroxyacyl-sphingosine (cer-EOS) has been shown to reduce the permeability of SC cell membranes and may be used to treat psoriasis scabs (Figure 1).

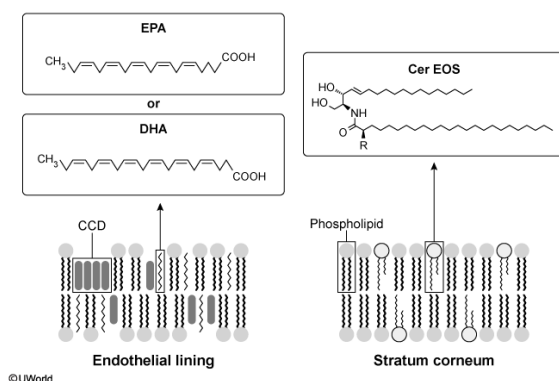


Figure 1 Organization of membranes in endothelial and epidermal cells

Treatment of endothelial cells with DHA has been observed to increase the fluidity of cell membranes. What structural feature of DHA, shown in Figure 1, most likely accounts for the increase in cell membrane fluidity?

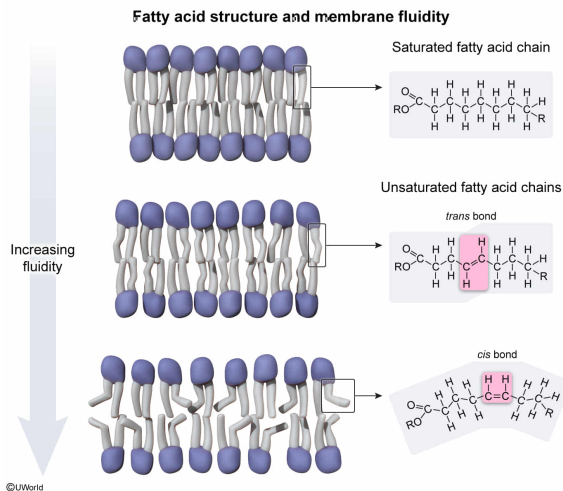
- ✓ ☒ A. The *cis* double bonds [80%]
☐ B. The *trans* double bonds [10%]

- ☐ C. The carboxyl groups [2%]
- ☐ D. The length of the carbon chain [6%]

Correct

80% answered correctly

Explanation:



Fatty acids are nonpolar molecules composed of unbranched hydrocarbon chains with carboxyl groups at one end. Humans synthesize fatty acids with an even number of carbon atoms, and chains usually range from 14 to 18 carbon atoms long. Fatty acids with no carbon-carbon double bonds are described as **saturated** because each carbon atom has the maximum number of hydrogen atoms possible. In contrast, an **unsaturated fatty acid** contains one (monounsaturated) or multiple (polyunsaturated) double bonds that can be in either the *cis* or *trans* configuration.

Unsaturation contributes significantly to membrane fluidity. The carbon-carbon double bonds decrease the melting temperature of fatty acid chains and increase the average space between lipids. As a result, lipids with unsaturated fatty acid chains are more fluid at a given temperature than saturated lipids of the same length. The **cis configuration** is particularly important in cell membranes as it introduces a bend or "kink" in the fatty acid chain that prevents phospholipids from stacking together and interacting efficiently. Figure 1 shows that DHA contains multiple *cis* bonds, which likely account for the observed increase in membrane fluidity when cells are treated with DHA.

(Choice B) Figure 1 shows that DHA does not contain any *trans* bonds.

(Choice C) The carboxyl groups of free fatty acids such as DHA provide a hydrophilic region that can interact with the aqueous environment of the cell, but they do not contribute to membrane fluidity.

(Choice D) DHA contains a longer carbon chain than the average fatty acid found in human cells. Longer carbon chains generally lead to *reduced* membrane fluidity compared to shorter chains with similar bond configurations.

Educational objective:

Saturated fatty acids have no carbon-carbon double bonds, and therefore stack closely in the cell membrane. Unsaturated fatty acids have one or more double bonds in either the *cis* or the *trans* configuration. In particular, the *cis* configuration enhances membrane fluidity by introducing a bend or "kink" in the fatty acid chain.

Time spent:0 sec**Subject:**

Biochemistry

QID:401077**System:**

Carbohydrates, Nucleotides, and Lipids

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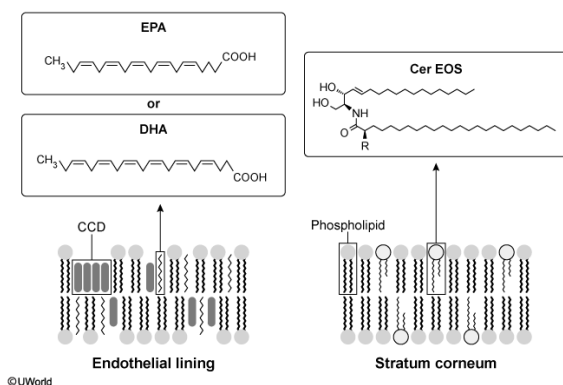


Figure 1 Organization of membranes in endothelial and epidermal cells

Phospholipids in the cell membrane could be separated experimentally according to their differences in:

- I. charge.
- II. mass.

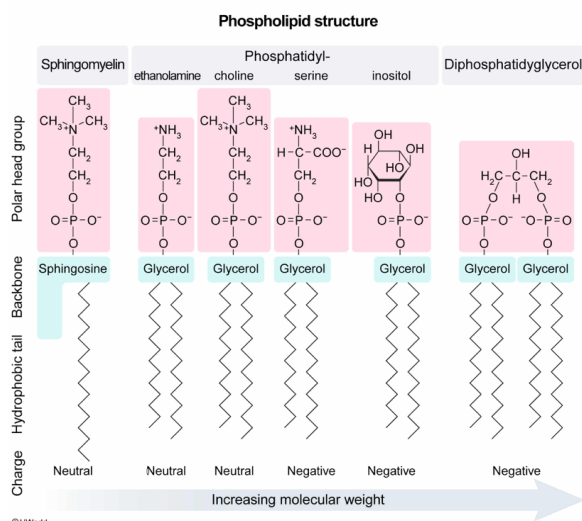
III. solubility.

- ☐ A. I only [2%]
☐ B. III only [18%]
☐ C. I and II only [29%]
✔ ☒ D. I, II, and III [48%]

Correct

48% answered correctly

Explanation:



Phospholipids (phosphatides) are the main structural lipids of the cell membrane. They are composed of a **hydrophilic polar head group** that contains a phosphate and a **hydrophobic tail** with one or two fatty acid chains attached to a carbon backbone. Each phospholipid has a unique mass, charge, and solubility due to distinct features in the backbone, polar head groups, and fatty acid chains. They may be separated based on each of the following properties:

- The **charge** on phospholipids is determined by the **structure of the polar heads**. The phosphate component of the head group may be linked to various chemical groups. Positively charged groups such as choline or ethanolamine neutralize the negative charge of the phosphate whereas neutral groups such as serine, inositol, or glycerol allow the head to maintain a net negative charge. Neutral and negative phospholipids can be separated by chromatography techniques such as high-performance liquid chromatography or ion exchange chromatography (**Number I**).
- The **mass** of phospholipids is established by the molecular weight of the backbone and the polar head groups. **Phospholipid backbones** can contain either a **glycerol** with two fatty acid tails or **sphingosine** with one fatty acid tail. Glycerophospholipids tend to be larger than **sphingophospholipids** because they have two fatty acid tails instead of one. Polar head groups may be as small as a phosphate (phosphatidic acid) or as large as phosphatidylglycerol. They can be

separated by [size-exclusion chromatography](#) (Number II).

- Phospholipid **solubility** depends on both the hydrophobic ("water-fearing") and hydrophilic ("water-loving") regions of the molecule. Charged head groups are more soluble than neutral groups because they more readily form hydrogen bonds whereas lipids with long hydrophobic chains have decreased solubility.

Phospholipids with different solubilities can be separated by [thin-layer chromatography](#) (Number III).

Educational objective:

Phospholipids are composed of a hydrophilic head group that contains a phosphate and a hydrophobic tail with one or two fatty acid chains. Phospholipids have either a glycerol or a sphingosine backbone. The composition of the head groups and the length of the tails create lipids with distinct charge, mass, and solubility.

Time spent:0 sec

Subject:

Biochemistry

QID:401078

System:

Carbohydrates, Nucleotides, and Lipids

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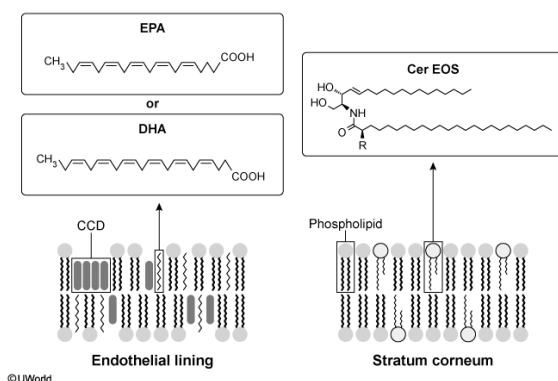


Figure 1 Organization of membranes in endothelial and epidermal cells

Prolonged starvation in patients diagnosed with hypertriglyceridemia would most likely promote:

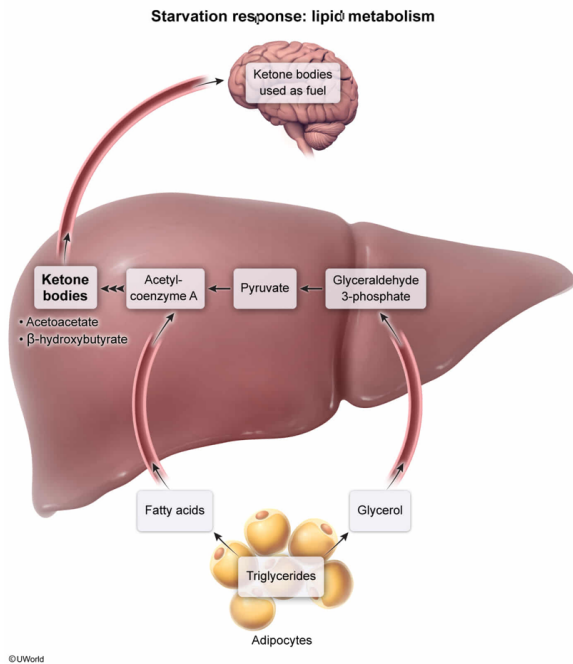
- ☐ A. the accumulation of glycerol in the bloodstream. [32%]
- ☒ B. the utilization of ketone bodies by the brain. [60%]
- ☐ C. the production of glycogen stores in the liver. [5%]

☐ D. the synthesis of amino acids in the liver. [2%]

Correct

60% answered correctly

Explanation:



Starvation causes a significant decrease in blood glucose levels and stimulates the release of the hormone glucagon, which signals for the release of glucose from glycogen stores (glycogenolysis). To preserve glucose for the tissues that require them, the liver also begins to break down triacylglycerol reserves. **Triacylglycerols** are esters of glycerol with three fatty acid chains that function as the main energy reserves in animals.

These **storage lipids** are formed in fat cells, or adipocytes, and yield the most ATP when broken down in the liver to glycerol and free fatty acids. Glycerol is rapidly **converted** to glyceraldehyde 3-phosphate for glucose synthesis (gluconeogenesis) while free fatty acids enter the mitochondria, where they are oxidized to generate acetyl-CoA for the citric acid cycle to produce energy. Excess **acetyl-CoA** that accumulates beyond what is needed for the liver can be converted to **ketone bodies** such as acetoacetate and 3-hydroxybutyrate, which are then exported into the circulation. Although the brain primarily uses glucose as a fuel source, it can also use ketone bodies as an energy source when needed.

Based on the passage, patients with hypertriglyceridemia have a surplus of triacylglycerol in their bloodstream. During starvation, excess triacylglycerol in the bloodstream and liver is ultimately converted to ketone bodies in these patients and utilized by the brain and other tissues.

(Choice A) Glycerol is rapidly converted to glycerol 3-phosphate and then oxidized to glyceraldehyde 3-phosphate during prolonged starvation. It does not accumulate.

(Choice C) Glycogen stores are depleted within 12–24 hours of fasting and are not available after prolonged starvation.

(Choice D) Prolonged starvation causes amino acids in proteins to be broken down to synthesize glucose.

Educational objective:

Triacylglycerols, also known as triglycerides, are storage lipids that consist of three fatty acid chains attached to a glycerol molecule. During prolonged starvation, acetyl-CoA produced from fatty acid oxidation in the mitochondria is converted to alternative fuel molecules known as ketone bodies.

Time spent:0 sec

Subject:
Biochemistry

QID:401079

System:
Carbohydrates, Nucleotides, and Lipids

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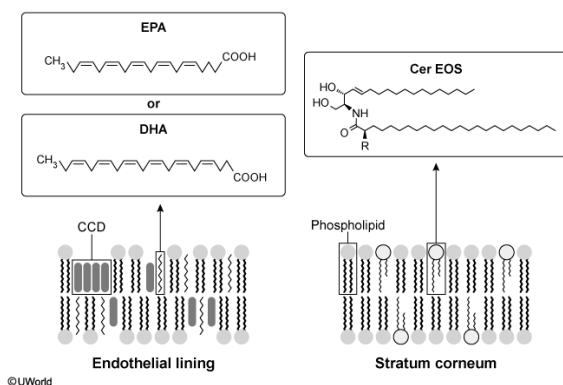


Figure 1 Organization of membranes in endothelial and epidermal cells

Based on the passage, individuals with atherosclerosis or psoriasis would benefit from therapies that result in which of the following changes, respectively?

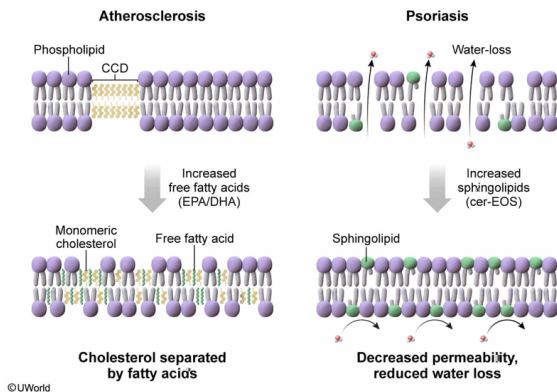
- ☐ A. A decrease in LDL or a decrease in ceramides [6%]
- ☐ B. A decrease in glycolipids or an increase in cholesterol [3%]
- ☒ C. An increase in free omega-3 fatty acids or an increase in sphingolipids [86%]

☐ D. An increase in terpenes or a decrease in saturated lipids [2%]

Correct

86% answered correctly

Explanation:



Membrane fluidity is determined partially by the concentration of cholesterol and the tail length of fatty acids. **Cholesterol** is a steroid alcohol that maintains membrane fluidity at an optimal level. At **high temperatures**, it provides rigidity and stabilizes the membrane; at low temperatures, it increases fluidity and prevents the membrane from solidifying. **Short unsaturated fatty acid tails** increase membrane fluidity by preventing phospholipids from clustering together, but **longer saturated tails** such as those typically found in sphingolipids induce lipid clustering and decrease fluidity. As described in the passage, changes in cholesterol levels and sphingolipid concentration can lead to diseases such as atherosclerosis and psoriasis, respectively.

Atherosclerosis results from the accumulation of cholesterol into CCDs in the membranes of endothelial cells. According to the passage, omega-3 fatty acids can inhibit CCD formation by separating cholesterol molecules. Therefore, an increase in free omega-3 fatty acids would likely help treat atherosclerosis.

Psoriasis is marked by an increase in membrane permeability due to a lack of ceramides in the cell membranes of the SC. The long fatty acid chains in sphingolipids such as cer-EOS cause lipids to cluster and decrease permeability. Therefore, increased sphingolipid levels could help treat psoriasis.

(Choice A) Decreasing LDL would improve atherosclerosis but ceramides should increase, not decrease, in psoriasis therapies.

(Choice B) Glycolipids are not involved in atherosclerosis, so decreasing their levels would not affect atherosclerosis. According to the passage, ceramides (a type of lipid derived from **sphingosine**), not cholesterol, are needed to prevent psoriasis.

(Choice D) Terpenes are precursors in cholesterol synthesis, so increased terpene concentration could worsen atherosclerosis by increasing cholesterol production. Saturated lipids such as cer-EOS are necessary for psoriasis treatments.

Educational objective:

Cholesterol levels and fatty acid tail length help control membrane fluidity by regulating rigidity and lipid clustering. Changes in levels of either can result in disease.

Time spent:0 sec**Subject:**

Biochemistry

QID:401080**System:**

Carbohydrates, Nucleotides, and Lipids

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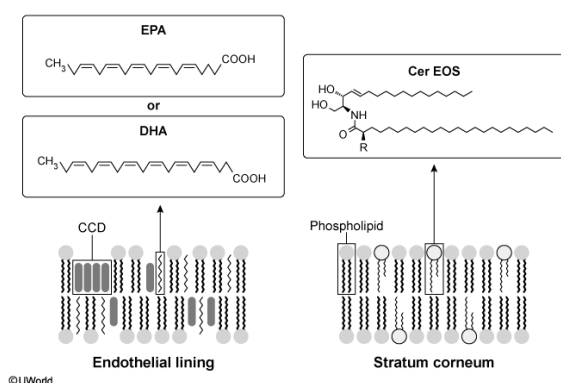


Figure 1 Organization of membranes in endothelial and epidermal cells

Structural lipids in the cell membrane can be derived directly from dietary sources or from lipase-mediated catabolism. Which statement correctly describes lipid processing during digestion?

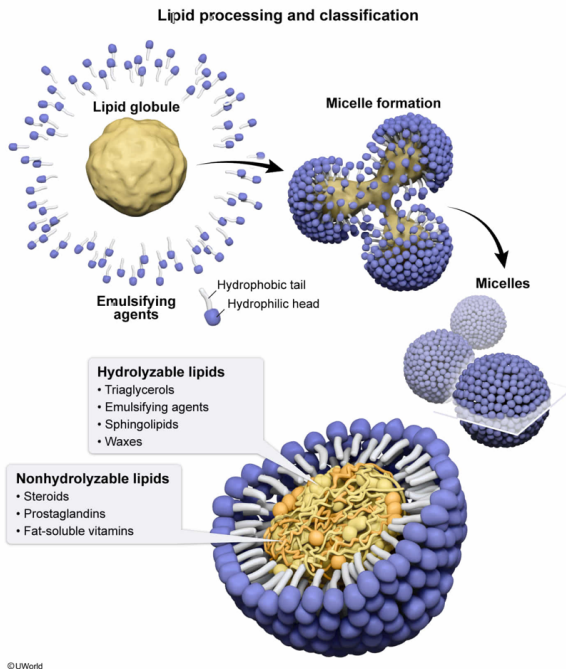
- ☐ A. All lipids are emulsified, but waxes cannot be hydrolyzed. [14%]
- ☐ B. Phospholipids can be hydrolyzed, but only steroids and vitamins are emulsified. [7%]

- ☐ C. Triglycerides are hydrolyzed, but only prostaglandins can be emulsified. [3%]
- ✓ ☒ D. All lipids can be emulsified, and triglycerides and phospholipids are hydrolyzed. [74%]

Correct

74% answered correctly

Explanation:



Lipid processing begins in the small intestine (duodenum), where bile salts break down lipid globules into smaller droplets in a process called **emulsification**. This process results in the formation of spherical structures known as **micelles**, which are composed of a hydrophobic core containing the nonpolar hydrocarbon tails of lipids and an outer shell of polar head groups that make contact with water. The formation of micelles increases the surface area of lipids available for **hydrolysis by lipases**.

Lipases are enzymes that digest certain emulsified lipids to facilitate their absorption, although some lipids are nonhydrolyzable. **Hydrolyzable lipids** contain ester bonds that can be cleaved by lipases through the addition of a water molecule (hydrolysis). These lipids include triacylglycerols, phospholipids, sphingolipids, and waxes. **Nonhydrolyzable lipids** do not contain the ester linkages necessary for lipase digestion. The most common nonhydrolyzable dietary lipids are cholesterol (steroids) and fat-soluble vitamins (A, D, E, and K).

The question states that some dietary lipids are catabolized (ie, broken down) by lipases whereas others can be absorbed directly. The most accurate description of lipid processing acknowledges that all lipids are emulsified during digestion but only some lipids, including triglycerides and phospholipids, contain hydrolyzable ester bonds.

(Choice A) **Waxes** are hydrolyzable lipids that consist of fatty acid chains bound to long-chain alcohols by an ester bond.

(Choices B and C) All dietary lipids are first emulsified in the aqueous environment of the small intestine. **Prostaglandins** are nonhydrolyzable lipids that function as signaling molecules.

Educational objective:

Emulsification increases the surface area of lipids by breaking down large globules into spherical structures called micelles. Micelles have a hydrophobic core, which contains the nonpolar hydrocarbon tails of lipids, and an outer shell of polar head groups, which make contact with water. Lipases can cleave the ester bonds in hydrolyzable lipids such as triglycerides, phospholipids, and waxes by adding a water molecule (ie, hydrolysis).

Time spent:0 sec

Subject:
Biochemistry

QID:401081

System:
Carbohydrates, Nucleotides, and Lipids

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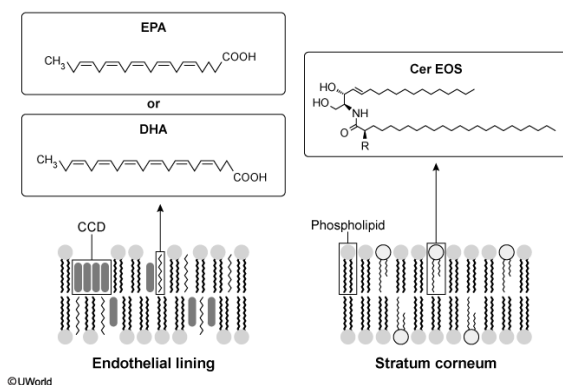


Figure 1 Organization of membranes in endothelial and epidermal cells

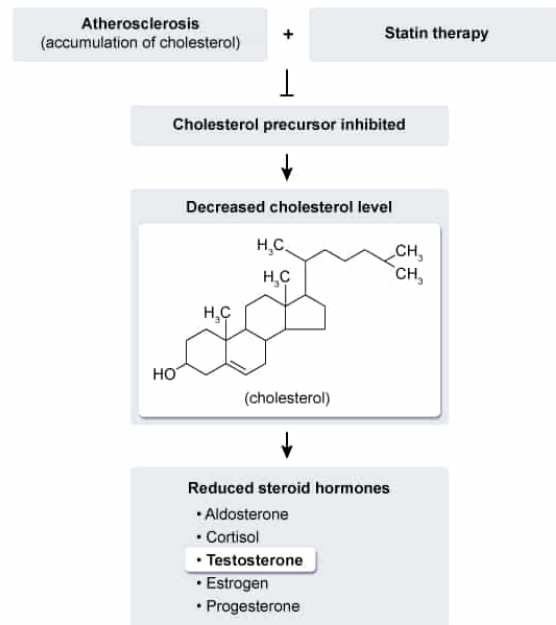
Individuals undergoing statin therapy would most likely exhibit:

- ☐ A. high levels of leptin. [11%]
- ☒ B. low levels of testosterone. [69%]
- ☐ C. low levels of insulin. [11%]
- ☐ D. high levels of cortisol. [8%]

Correct

69% answered correctly

Explanation:



Steroids are tetracyclic derivatives of terpenes that contain **four hydrocarbon rings** fused together (ie, three 6-membered rings and one 5-membered ring). **Cholesterol** is a steroid alcohol and the precursor molecule for the synthesis of bile salts, vitamin D, and steroid hormones. Cholesterol and its hormone derivatives are classified as **amphipathic molecules** because they contain a hydrophilic region and a hydrophobic region, which allows them to diffuse across cell membranes.

Common steroid hormones include testosterone, cortisol, aldosterone, estrogen, and progesterone, all of which are secreted by endocrine glands into the bloodstream where they travel to their target cells. According to the passage, statin therapy treats atherosclerosis by inhibiting the production of a cholesterol precursor. Therefore, statins may decrease the levels of cholesterol and consequently the levels of steroid hormones such as testosterone.

(Choices A and C) Peptide hormones such as leptin and insulin are made from chains of amino acids. Neither would be directly affected by the reductions in cholesterol synthesis caused by statin therapy.

(Choice D) Cortisol concentration would be *reduced* in individuals undergoing statin therapy because cholesterol is necessary for cortisol production.

Educational objective:

Cholesterol is a steroid alcohol and the precursor molecule for the synthesis of steroid hormones, bile salts, and vitamin D. Common steroid hormones include cortisol, aldosterone, estrogen, testosterone, and progesterone. Inhibition of cholesterol synthesis reduces the levels of steroid hormones.

Time spent:0 sec

Subject:
Biochemistry

QID:401082

System:
Carbohydrates, Nucleotides, and Lipids